

Jinjin Zhao

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About

Research Interests: AI Agents (LLM) Orchestration, Data Science Governance, Machine Learning Infrastructure, Data Platforms, Databases

Languages: Python, SQL, C

Skills: PyTorch, scikit-learn, Pandas, Airflow, Git, Unix, Amazon Web Services, Google Cloud Platform

Education

PhD University of Chicago, Computer Science Sept. 2019 to 2025 (est.)

- Advisor: Sanjay Krishnan (ChiData Database Group)
- Completed M.S. degree as a part of Ph.D. program

BSE Princeton University, Computer Science Sept. 2015 to May 2019

- Summa Cum Laude
- Minor in Statistics and Machine Learning

Selected Projects

TableVault: Contextual End-to-End Management of LLM Data Pipelines ([Website](#)) (On-going)

- A file-based open-source project to manage data tables and artifacts in complex and dynamic data workflows.
- Extended database transaction techniques, such as two-phase locking and write ahead logs, to enable reverts, restarts, and pauses on all write executions.
- Tracked data dependencies between tables by enforcing YAML configurations for Python executions.
- Enabled governed human-LLM collaboration, capturing data edits from both practitioners and agents.

DLog: A Compressed Query and Storage Framework for Fine-Grained Array Lineage

- Augmented the Numpy library to record cell-to-cell operational lineage efficiently with memory optimizations.
- Introduced a new range-based compression algorithm that improves storage space and query time of the resulting lineage graph by up to 2000x and 20x respectively.

Tracing Variation in Data Science Workflows with Jupyter Notebook Logging

- A tool for Jupyter Notebooks and Python to log execution traces of data science assignments at University of Chicago.
- Analyzed the traces to capture user variation trends in data science usage (e.g. most errors are resolved within 1-2 code executions).
- Validated some common conceptions in data science (e.g. data cleaning takes about 80% of the work).

Experience

Linea Labs, Research Intern

CA, USA

June 2023 to Sept. 2023

- Co-designed the initial MVP for Airflow pipeline reproducibility with a 6-person startup team, owning research and implementation of core lineage-tracking and LLM-based code-analysis features.

Princeton Plasma Physics Lab, Research Intern

NJ, USA

June 2018 to July 2018

- Compiled data on two years of DIII-D tokamak experiments and trained neural network models to predict pedestal features driving fusion output.

Meta, Software Engineer Intern

- Developed a full-stack Hack (PHP) and MySQL tool for storing and retrieving test artifacts (e.g., logs and build files) from internal pre-commit automated runs.
- Scaled to over 200 million files per week, supporting the majority of internal code development.

WA, USA
June 2017 to Aug. 2017

Meta, Facebook University Intern

- Designed and built an independent Android app with an internal music player that generated Spotify playlists based on nearby concerts.

CA, USA
June 2016 to Aug. 2016

Publications

Metadata Management for AI-Augmented Data Workflows 2025

Jinjin Zhao, Sanjay Krishnan

Under Submission [Paper](#) 

Fast Capture of Cell-Level Provenance in Numpy 2025

Jinjin Zhao, Sanjay Krishnan

ProvenanceWeek@SIGMOD [Paper](#) 

TableVault: Managing Dynamic Data Collections for LLM-Augmented Workflows 2025

Jinjin Zhao, Sanjay Krishnan

NOVAS@SIGMOD [Paper](#) 

Learning Lineage Constraints for Data Science Operations 2025

Jinjin Zhao

arXiv [Paper](#) 

Quantifying Variation in Data Science Workflows with Fine-Grained Procedural Logging. 2024

Jinjin Zhao, Avidgor Gal, Sanjay Krishnan

Under Submission [Paper](#) 

Compression and In-Situ Query Processing for Fine-Grained Array Lineage. 2024

Jinjin Zhao, Sanjay Krishnan

ICDE [Paper](#) 

Data Makes Better Data Scientists. 2023

Jinjin Zhao, Avidgor Gal, Sanjay Krishnan

HILDA@SIGMOD [Paper](#) 

AMIR: Active Multimodal Interaction Recognition from Video and Network Traffic in Connected Environments. 2023

Shinan Liu, Tarun Mangla, Ted Shaowang, **Jinjin Zhao**, Sanjay Krishnan, Nick Feamster
UbiComp/IMWUT [Paper](#) 

Data Station: Delegated, Trustworthy, and Auditable Computation to Enable Data-Sharing Consortia with a Data Escrow. 2022

Siyuan Xia, Zhiru Zhu, Chris Zhu, **Jinjin Zhao**, Kyle Chard, Aaron J. Elmore, Ian Foster, Michael Franklin, Sanjay Krishnan, Raul Castro Fernandez

VLDB [Paper](#) 

Towards Causal Query Answering for Debugging Video Analytics Systems. 2022

Ted Shaowang*, **Jinjin Zhao***, Stavos Sintos, Sanjay Krishnan

HILDA@SIGMOD [Paper](#) 

Prediction of DIII-D Pedestal Structure From Externally Controllable Parameters. 2021

Emi Zeger, Florian Laggner, Alessandro Bortolon, Cristina Rea, Orso Meneghini, Samuli Saarelma, Brian Sammuli, Sterling Smith, **Jinjin Zhao**

IEEE Transactions on Plasma Science [Paper](#) 

Activities And Awards

- **2018 - 2024 Teaching Assistant:** COS 397/497 Fall 2018 (*Princeton University*), CMSC 16100 Autumn 2019 (*University of Chicago*), CMSC 21800 Autumn 2020/2023 (*University of Chicago*), DATA 13600 Spring 2024 (*University of Chicago*)
- ICDE'2024 Travel Award, NSF
- 2022 University Unrestricted Fellowship, University of Chicago
- 2020 - 2022 Curriculum and Social Minister, UChicago CS Graduate Student Ministry
- 2020 Lab Coordinator, CDAC (high school data science research summer program)
- OSDI'2020 Diversity Grant, USENIX Association
- 2019 Neubauer Graduate Scholarship, University of Chicago
- 2016 YHacks 1&1 Prize Winner